

PROTOCOL FOR SINGLE CELL SEEDING OF BBB STEM CELL MODEL

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Updated by Emma Hollmann 10/21/16

Updated by Emma Hollmann 6/5/17 to reflect Countess II use

MATERIALS:

1. Plasticware

Fisher

Corning tissue culture plates

15 mL conical tubes

Microfuge tubes

ThermoFisher

Countess Cell counting chamber slides

2. Equipment

ThermoFisher

Countess II automated cell counter

3. Reagents

Thermo Fisher Scientific

Accutase

A1110501 100 mL

Ultrapure distilled water

10977023 500 mL

PBS

14190-144 500 mL

Tocris (R&D Systems)

ROCK inhibitor (Y-27632 dihydrochloride)

1254

10 mg

E8 media prepared in-house

REAGENT PREP:

ROCK inhibitor:

Make 10mM working stock solution by diluting 10mg ROCK inhibitor into 3.12 mL ultrapure distilled water. Use at 1:1000 for 10 μ M final concentration. Aliquots can be stored long term at -80°C for up to 1 year and frozen/thawed as many times as necessary.

PROCEDURE:

1. Seeding cells for BBB differentiation using single cell seeding (Day -1)

- Stem cells should be at appropriate splitting density; that is, the colonies should just be starting to touch each other.
 - Matrigel plates should be ready to use
 - Procedure was optimized using IMR90-4 pluripotent stem cells. The procedure has successfully been extended to CC3, CD12, SM14, and DSP-mEGFP pluripotent stem cells.
1. Manually transfer spent medium to a 15 mL conical. Save 1 mL of media for every well being passaged.
 2. Wash each well once with 2 mL PBS.
 3. Add 1 mL accutase (warmed to room temperature) to each well.
 4. Incubate at 37 °C until the cells are beginning to detach (approx. 3-5 min)
 5. Using p1000, collect cells, and spray gently over surface 2-3x to dislodge any remaining cells. Pipetting more than this will reduce cell viability.
 6. Collect cell in the 15 mL conical containing spent medium.
 7. Spin down cells for 4-5 min at 1000 rpm.
 8. Aspirate media, resuspend cells in 1 mL E8 medium. Thoroughly triturate 2-3 times using p1000 to yield single cell suspension.
 9. Take 10 uL of cells to count, drawing from the middle of the sample to prevent bias from settling cells. Transfer these cells to a clean microfuge tube.
 10. Dilute the 10 uL of cells in 10 uL of 0.4% Trypan blue. Transfer 10 uL of this diluted suspension to a Countess cell counting chamber; allow to sit for 30 seconds. **Note that this step is performed outside the hood and is NONSTERILE.**
 11. Insert the slide into the Countess II automated cell counter. Note the calculated live cell density. You will use this density (not the total density) to calculate the number of cells needed for seeding.
 12. Calculate appropriate volume of cells to add to each 6-well.
 - a. Typical seeding number for IMR90-4 iPSCs is between 100,000 – 120,000 cells per well.
 - b. Typical seeding number for CC3 and CD12 iPSCs is 150,000 cells per well.
 13. Resuspend cells in appropriate volume of E8 + 10 uM ROCK inhibitor (2 mL/well). If you have excess cells, discard the excess before resuspending.
 14. Place plate in incubator, and shake plate quickly back and forth (not swirling) to distribute cells evenly.
 15. Approximately 24 hours later (Day 0), initiate differentiation.